Useful Moodle Plugins *for* Programming Language Instructors

This article introduces the reader to the Moodle plugins, CodeRunner and Virtual Programming Lab (VPL).

A learning management system is a tool used to create and manage educational and training programs. Moodle is a very popular learning management system; in fact, it is so popular that people may not even know of an alternative to it, off hand. Moodle is free and open source software licensed under the GNU General Purpose License. It has been developed using PHP and it can be used to create additional study material for academic subjects as well as to enable blended learning. The term 'subject' might be a bit misleading. In many parts of the world, like in Europe or the USA, the term used to refer to the same unit of study is 'course', which, in the Indian context means a collection of subjects. For example, in our country, a degree course in physics might have a subject called astrophysics.

Blended learning is an educational strategy in which traditional classroom methods are combined with online digital media to make the learning process more effective. An example of blended learning is a flipped classroom where the instructional materials are provided outside the classroom as online resources and discussions, while traditional homework is done inside the classroom. The arrival of Moodle has made the creation of course material that follows the blended learning model very easy.

In this article, we will focus on a lesser-known feature of Moodle called plugins. Since there are a large number of plugins available for use with Moodle, let's discuss just two that are extremely useful to academicians who work as programming language instructors. These are CodeRunner and Virtual Programming Lab (VPL). Both allow the instructor to evaluate programming assignments very easily. CodeRunner is relatively simple and is available as a question type. VPL is a more complicated plugin, which offers features like interactive program editing, reviewing, plagiarism checking, etc.

Installing Moodle

The first step is to have a Web server solution stack package installed in your system. We are using XAMPP in our systems, so let us look at how it can be used to configure and run Moodle—but remember that you can use any other Web server of your choice to get the same effects. XAMPP is free and open source software consisting of the Apache HTTP server, MariaDB database, and interpreters for PHP and Perl. The installation of XAMPP is straightforward and after completing it, execute the command *sudo /opt/lampp/lampp start* on a terminal to start the XAMPP server. Now open a